# 16. Worked example



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# Worked example: matrix exponential

▶ 11:54 / 11:54 ▸ 2.0x

But basically, the key point here

was just to remember where is the matrix exponential coming

from, basically, from the eigenvalues and eigenvectors

of the original matrix present in the system,

and where is the definition coming from,

why do we define it as phi of t phi minus 1 of 0,

and how to use it then to give the solution

to an initial value problem.

**So that ends this recitation.**

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## Review

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Which of the following matrices could be an exponential matrix of the form $e^{\mathbf{A}t}$?

(Choose all that apply.)

- [x] $\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}$
- [ ] $\begin{pmatrix} e^t & e^{2t} \\ -2e^t & -2e^{2t} \end{pmatrix}$
- [ ] $\begin{pmatrix} e^{-3t} & e^t & te^t \\ 2e^{-3t} & 0 & 0 \\ 0 & e^t & e^t \end{pmatrix}$
- [ ] $\begin{pmatrix} e^{-3t} & e^t & e^t \\ 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix}$
- [ ] $\begin{pmatrix} e^{it} & e^{-it} \\ -ie^{it} & ie^{-it} \end{pmatrix}$

✔

**Solution:**

We need to check that $\mathbf{X}(0) = \mathbf{I}$. We identified $\mathbf{X}(0)$ in a previous concept check:

- $\left.\begin{pmatrix} \cos(t) & \sin(t) \\ -\sin(t) & \cos(t) \end{pmatrix}\right|_{t=0} = I$
- $\left.\begin{pmatrix} e^t & e^{2t} \\ -2e^t & -2e^{2t} \end{pmatrix}\right|_{t=0} = \begin{pmatrix} 1 & 1 \\ -2 & -2 \end{pmatrix}$
- $\left.\begin{pmatrix} e^{-3t} & e^t & te^t \\ 2e^{-3t} & 0 & 0 \\ 0 & e^t & e^t \end{pmatrix}\right|_{t=0} = \begin{pmatrix} 1 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$

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- $\begin{pmatrix} 0 & e^t & 0 \\ 0 & 0 & e^t \end{pmatrix}\Bigg|_{t=0} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- $\begin{pmatrix} e^{it} & e^{-it} \\ -ie^{it} & ie^{-it} \end{pmatrix}\Bigg|_{t=0} = \begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$

The only potential exponential matrix is the first one.

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Answers are displayed within the problem

## 16. Worked example

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